

Can Historical Logs Reliably Evaluate a New Policy?

A reinforcement-learning case study of recency, coverage, and uncertainty in off-policy evaluation

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Submission note for final author review. Data Science Collective says it does not accept fully AI-generated stories and asks for clear, accessible writing without unnecessary jargon. Before submission, this draft should receive a final personal pass: verify the numerical claims against the experiment, rewrite any sentence that does not sound like your own voice, and add or remove interpretation until you can stand behind every claim. The goal is genuine authorship and clarity, not AI-detector evasion.¹

When historical logs become fragile

A new policy can be easy to fit and surprisingly hard to trust. The problem is not limited to reinforcement learning. Recommendation systems, ad platforms, operational decision tools, and other data products all learn from logs created by older rules. If the environment changes, those logs may still be large while becoming less useful for the exact policy we now want to evaluate.

Off-policy evaluation (OPE) is designed for this setting. It estimates the value of a target policy using data collected by another behavior policy. In practice, three questions quickly become entangled. Are the old observations still relevant? Did the behavior policy explore enough actions to support the new policy? And is the estimated improvement large enough relative to its uncertainty to justify a change?

This article uses a small reinforcement-learning laboratory to separate those questions. The numerical sequence happens to come from daily QQQ returns because it gives a long, externally observed time series and makes one-step counterfactual rewards easy to audit under a price-taking assumption. The point is methodological rather than financial. QQQ is a test bed for logged-policy evaluation, not the basis of a trading recommendation.

The results led to a simple framing that I found more useful than asking which estimator “won”: evidence quality under change has at least three parts—**relevance**, **coverage**, and **precision**. They fail for different reasons.

A hidden check makes the experiment measurable

The setup is a one-step contextual bandit. At each decision point, the learner receives a compact context made from recent returns, volatility, drawdown, a moving-average signal, and current exposure. It then chooses one of three actions: reduce exposure, hold, or increase exposure.

Only the selected action’s reward is shown to the learner. That preserves ordinary bandit feedback. After the next return becomes known, however, the immediate reward of all three actions can be reconstructed at the same pre-action position. The two unchosen rewards remain hidden during learning and OPE, then become available for scoring.

That hidden reward vector acts as a one-step oracle. It is deliberately narrow. A different action today would change the position carried forward, so the oracle does not reconstruct an entire alternative path or a full Markov decision process. It answers a cleaner question: at this logged decision point, how good was each immediate action?

A stochastic linear Thompson-sampling policy generates the behavior log. Because the probability of the selected action is recorded, standard OPE methods can be applied: importance sampling (IPS), self-normalized IPS (SNIPS), direct modeling (DM), and doubly robust estimation (DR). The oracle lets us judge their estimates rather than merely compare estimators with one another.

Good value estimates can still select the wrong policy

The first benchmark evaluates four target policies: a long-history baseline, a recent-history candidate, a conservative mixture of those two, and a uniform reference policy.

Across three independently simulated behavior logs, IPS produced the lowest average value-estimation error, while DR had the smallest average signed bias. The mean absolute errors were 0.000455 for IPS, 0.000467 for SNIPS, 0.000550 for DR, and 0.001057 for DM.

Those numbers are reasonably small, but policy selection was much weaker. Best-policy accuracy stayed around 35–40% across the estimators. A method could therefore estimate values fairly closely and still choose the wrong candidate when several policies were near one another. For an applied system, that comparison is usually what matters: not just “What is the value?” but “Is the candidate better than the baseline by enough to justify changing the policy?”

Effective sample size (ESS) explained part of the error. ESS is a way to summarize how many observations remain effectively useful after importance weighting. Higher ESS was generally associated with lower OPE error, but the relationship was far from exact.

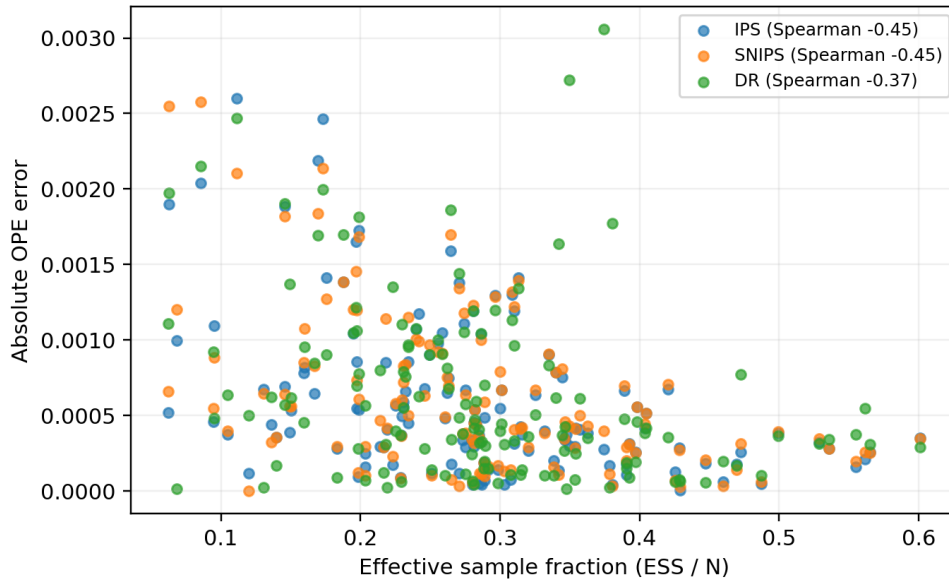


Figure 1: Effective sample fraction and absolute OPE error. Better target-policy overlap usually helped, but ESS did not explain all of the variation.

That gap is important. ESS can warn that a target policy is poorly supported by the historical log. It cannot tell us whether those observations are stale, whether the reward model is misspecified, or whether a tiny policy-value difference is buried in statistical noise.

Recent data is not automatically better

Nonstationarity often suggests a simple fix: use only the newest observations. I tested that idea by holding the DR evaluator fixed and changing only the historical evidence supplied to it.

The alternatives included all available history, trailing windows of 252, 126, and 60 observations, exponential time decay, and time decay combined with a simple regime-similarity weight. Across the three behavior logs, the short fixed windows performed much worse than the longer-memory rules.

This is not an argument for keeping every old observation forever. One robustness seed favored exponential decay, and selective forgetting helped in some smaller regime-specific slices. The more useful point is that freshness has a cost.

Historical evidence	Mean OPE MAE
All history	0.000519
Time decay	0.000568
Decay + regime similarity	0.000582
Last 252 observations	0.000914
Last 126 observations	0.001415
Last 60 observations	0.001587

Table 1: Average OPE error under different historical-memory rules.

Dropping old data may reduce stale-distribution bias, but it also removes sample size and action support. In this study, aggressive forgetting often lost more useful information than it gained in recency.

So the memory problem is not solved by choosing “old” or “new.” A long sample can lean toward an outdated regime, while a short sample can become too thin for stable counterfactual evaluation. The right memory horizon depends on both change in the environment and the evidence lost by forgetting.

Recency and coverage are different problems

The most revealing part of the study was the separation between recent evidence and target-policy coverage.

Suppose the evaluator needs more current data. One option is to add recent observations generated by the old behavior policy. Another is to perform **simulated target-aware re-logging** on those same historical dates, shifting some logging probability toward actions favored by the candidate policy.

Both interventions use recent dates, but they add different information. Recent old-policy observations mainly improve environmental recency. Target-aware re-logging mainly improves action coverage.

At 80 added observations, the cross-seed averages looked like this:

Evidence source	Mean MAE	Mean ESS
History only	0.000633	166.1
Target-aware re-logging	0.000605	201.4
Recent old-policy evidence	0.000578	159.2

Table 2: Recent evidence changed OPE accuracy and coverage in different ways.

Target-aware re-logging raised ESS substantially, which is what we would expect from better overlap with the candidate policy. Yet the recent old-policy observations achieved the larger reduction in OPE error even though ESS did not improve.

The practical lesson is not that one source of evidence dominates. A log can have strong target-policy coverage and still describe an outdated reward relationship. It can also be very recent while giving poor support to actions that the candidate policy prefers. One ESS threshold cannot tell those cases apart.

Confidence can become abstention

A positive OPE estimate is not yet a policy decision. What matters is the estimated difference between the candidate and the baseline,

$$\widehat{\Delta V} = \widehat{V}_{\text{DR}}(\pi_{\text{new}}) - \widehat{V}_{\text{DR}}(\pi_{\text{base}}). \quad (1)$$

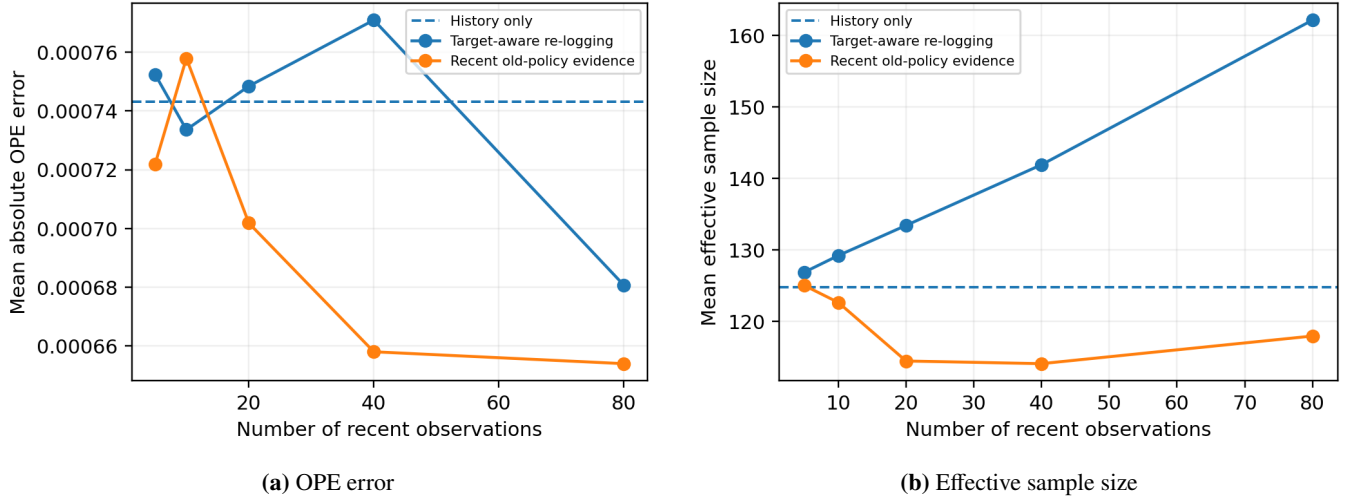


Figure 2: Adding recent observations can improve current relevance without improving overlap, while target-aware re-logging improves overlap more directly.

A moving-block bootstrap gives a standard error for that difference. A one-sided 95% lower confidence bound is

$$LCB = \widehat{\Delta V} - 1.645 \widehat{SE}(\widehat{\Delta V}). \quad (2)$$

The logic is simple: do not favor a candidate merely because the point estimate is slightly above zero. The estimated advantage should survive a reasonable allowance for uncertainty.

On the held-out QQQ windows, that rule was extremely conservative. In the main behavior seed, the median true candidate-baseline value gap was about 5.1×10^{-5} , while the median bootstrap standard error was about 4.2×10^{-4} . The uncertainty was roughly eight times larger than the typical signal.

A controlled synthetic stress test made the mechanism easier to see. The lower-confidence-bound rule suppressed weak or negative signals, then adopted more often as the true advantage became large enough relative to the noise.

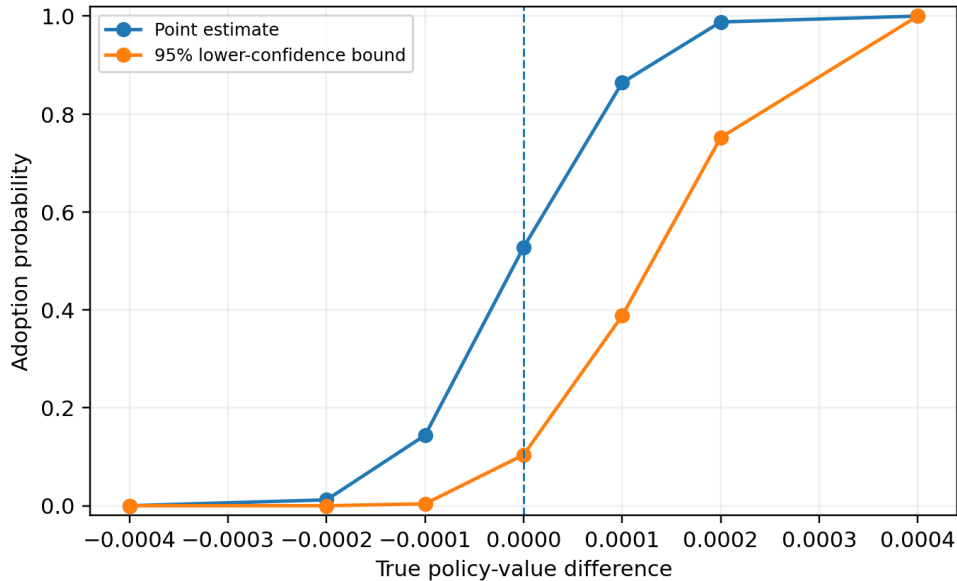


Figure 3: A confidence-based gate trades caution for power. Near zero it tends to abstain; larger true advantages are easier to detect.

Abstention should not automatically be counted as success. A gate that almost never approves an update may avoid some bad changes, but it cannot identify useful ones either. In a live decision system, repeated abstention is better interpreted as a request for more evidence, better coverage, or a stronger evaluator.

Exploration also builds future evidence

The last experiment changes the logging policy itself. The minimum action probability is varied from $\epsilon = 0.01$ to 0.10, and the logging-and-evaluation procedure is repeated 20 times at each setting.

Across that tested range, mean ESS rose from 49.8 to 81.1, an increase of about 63%. Mean absolute OPE error fell from 0.001202 to 0.000479, a reduction of about 60%.

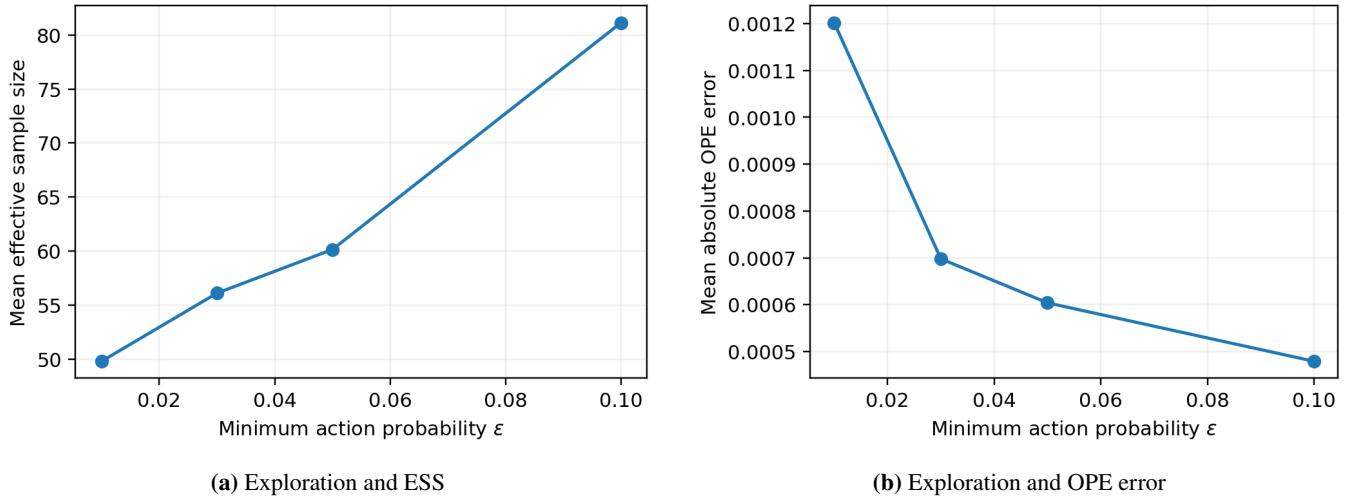


Figure 4: Within the tested range, broader logging coverage produced larger ESS and lower OPE error.

Nothing here proves that more exploration is always better or that $\epsilon = 0.10$ is optimal. The logger already had nontrivial coverage, only three actions were available, and different logged actions also change the simulated position path.

Still, the result gives exploration another role. We usually introduce exploration as a way to discover better actions now. OPE shows that it can also preserve the ability to evaluate different policies later. A behavior policy creates a data asset. If that log becomes nearly deterministic, future target policies may be hard to assess simply because the actions they prefer were rarely observed.

What I would carry into another data-science system

The study does not support a universal rule such as “always keep all history,” “collect exactly 80 recent observations,” or “use a particular exploration rate.” Those would be conclusions fitted too tightly to one laboratory.

A more durable interpretation is that policy evaluation under change depends on three different properties of evidence:

$$\mathcal{E}_t = (\text{relevance, coverage, precision}). \quad (3)$$

Relevance asks whether historical observations still describe the environment of interest. **Coverage** asks whether the behavior log supports actions favored by the target policy. **Precision** asks whether the candidate-baseline value difference is large enough relative to uncertainty to support a decision.

That suggests a division of work that is useful beyond this example:

$$\text{policy learner} \longrightarrow \text{evidence manager} \longrightarrow \text{OPE} \longrightarrow \text{update gate}. \quad (4)$$

A learner proposes a policy. An evidence-management layer decides which historical and current observations are credible enough for comparison. OPE estimates value from that evidence. The final gate asks whether the difference is resolved well enough to justify an update.

The same pattern can appear in recommender systems, advertising, allocation problems, and other logged decision systems. A new policy may be easy to train and still be difficult to evaluate from the data left behind by its predecessor.

Limits and the next step

This remains a one-step contextual-bandit laboratory. It does not reconstruct full counterfactual trajectories, and it is not a deployable financial system. The behavior logs are simulated, the action space is intentionally small, and the current DR implementation uses Ridge reward models without time-block cross-fitting. Validation also uses the hidden oracle to compare evidence rules, which is scientifically useful but unavailable in real deployment.

Several robustness checks reuse the same underlying sequential history even when the simulated behavior log changes. They measure sensitivity to logging randomness, not independence across environments. A stronger follow-up should test the same ideas on a second, non-financial logged-decision problem and, where possible, use non-overlapping evaluation windows.

The next methodological step is **adaptive evidence management**: choose memory, request additional data, and adjust evaluation thresholds from observable drift, overlap, and uncertainty signals rather than from an oracle. That would turn the current laboratory insight into something closer to a deployable decision pipeline.

Resources

- **Working paper:** [insert arXiv or SSRN link]
- **Code and reproducibility materials:** [insert GitHub link]
- Miroslav Dudík, John Langford, and Lihong Li (2011), *Doubly Robust Policy Evaluation and Learning*.
- Yu-Xiang Wang, Alekh Agarwal, and Miroslav Dudík (2017), *Optimal and Adaptive Off-policy Evaluation in Contextual Bandits*.
- Philip Thomas, Georgios Theodorou, and Mohammad Ghavamzadeh (2015), *High Confidence Policy Improvement*.

The shortest summary is also the one worth keeping: learning a candidate policy is only the first step. The harder question is whether the evidence available today is relevant, well covered, and precise enough to justify believing that the candidate is actually better.